

Lecture 9: Temperature and Heat – Tutorial Sheet

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Use the Master Procedure: identify, sketch, convert units, predict, select the model, execute, evaluate. Use kelvin whenever T is raised to a power. Problems A1 and A2 are solved in full; final answers to the others are at the end. Useful values: $c_{\text{water}} = 4186$, $c_{\text{Cu}} = 390$, $c_{\text{Al}} = 900 \text{ J/(kg K)}$; $\alpha_{\text{steel}} = 1.2 \times 10^{-5}$, $\alpha_{\text{Al}} = 2.4 \times 10^{-5} \text{ K}^{-1}$; $\sigma = 5.67 \times 10^{-8} \text{ W/(m}^2\text{K}^4)$.

A. Worked examples

A1. A metal ring on a rod

A steel rod of diameter 2.500 cm at 20°C must pass through an aluminium ring whose hole has diameter 2.490 cm at 20°C. To what temperature must the ring be heated (the rod stays at 20°C)?

Solution. The hole expands like a solid of aluminium: $d = d_0(1 + \alpha\Delta T)$. We need $2.500 = 2.490(1 + 2.4 \times 10^{-5} \Delta T)$, so $\Delta T = \frac{2.500/2.490 - 1}{2.4 \times 10^{-5}} = \frac{4.016 \times 10^{-3}}{2.4 \times 10^{-5}} = 167 \text{ K}$. The ring must reach $T = 20 + 167 \approx 190^\circ\text{C}$. *Check:* the hole grows by 0.010 cm, which is 0.40% of its diameter, about $167 \times 2.4 \times 10^{-5}$.

A2. Cooling a drink

A 0.250 kg glass of water at 25°C is cooled by dropping in an 0.040 kg aluminium block at -10°C (no phase change, ignore the glass). Find the final temperature.

Solution. $m_w c_w (T_f - 25) + m_{\text{Al}} c_{\text{Al}} (T_f + 10) = 0$: $(0.250)(4186)(T_f - 25) + (0.040)(900)(T_f + 10) = 0$, i.e. $1046.5(T_f - 25) + 36.0(T_f + 10) = 0$, so $T_f = \frac{26\,163 - 360}{1082.5} = 23.8^\circ\text{C}$. The result lies between -10 and 25°C and is close to 25°C because the water's heat capacity (1047 J/K) is much larger than the block's (36 J/K).

B. Practice

B1. Conversions

(a) Convert -40°C to $^\circ\text{F}$. (b) At what temperature do the Celsius and Fahrenheit scales agree? (c) Convert 20°C to kelvin and a change of 25°F to kelvin.

B2. A bridge

A steel bridge girder is 100.0 m long. What is its change in length between a winter night at -5°C and a summer afternoon at 40°C ?

B3. Thermal stress

A steel rod ($Y = 2.0 \times 10^{11} \text{ Pa}$) is held between two rigid walls at 20°C . Find the stress when it is heated to 70°C .

B4. Energy to heat

How much energy is needed to heat 2.0 kg of aluminium from 20°C to 120°C? How long does a 500 W heater take?

B5. Conduction

A glass window of area 1.5 m² and thickness 4.0 mm ($k = 0.80 \text{ W}/(\text{m K})$) separates a room at 22°C from outside at 2°C. Find the heat current. How much energy is lost in an hour?

B6. Radiation

A hot plate of area 0.040 m² ($e = 0.90$) is at 400°C in a room at 20°C. Find the power it radiates and the net power it loses.

B7. Challenge: the method is not given

A 0.50 kg iron pan ($c = 450 \text{ J}/(\text{kg K})$) at 200°C is dropped into 2.0 kg of water at 20°C. (a) Find the final temperature (assume no boiling). (b) The container, with walls of area 0.10 m² and thickness 2.0 cm ($k = 0.040 \text{ W}/(\text{m K})$), then loses energy to 20°C surroundings. Estimate the initial heat current and how long the water takes to cool by 1.0 K.

Answers

- B1 (a) -40°F; (b) -40; (c) 293.15 K; $\Delta T = 14 \text{ K}$
B2 $\Delta L = (1.2 \times 10^{-5})(100.0)(45) = 5.4 \times 10^{-2} \text{ m} = 54 \text{ mm}$
B3 $Y\alpha\Delta T = 1.2 \times 10^8 \text{ Pa}$
B4 $Q = 1.8 \times 10^5 \text{ J}$; $t = 360 \text{ s}$ (6.0 min)
B5 $H = kA\Delta T/L = 6.0 \times 10^3 \text{ W}$; $2.2 \times 10^7 \text{ J}$ per hour
B6 emitted $H = e\sigma AT^4 = 4.2 \times 10^2 \text{ W}$ ($T = 673 \text{ K}$); net $4.0 \times 10^2 \text{ W}$
B7 (a) $T_f = 24.7^\circ\text{C}$; (b) $\Delta T = 4.7 \text{ K}$, $H \approx 0.94 \text{ W}$; $t \approx 8.9 \times 10^3 \text{ s}$ (2.5 h)